w/ carry $\{$ lsb+1 := 1

$\forall$ bits $\leq$ lsb := 0 lsb-1

if $f(rax_0 = \{$ [- ... 1 0 0 0 0 ... 0] msb, [1 1 1 1 1 0 0 0 0] lsb $\})$ = [0 ... 0 1 0 0 ... 0] msb, [1 1 1 1 1 0 0 0 0] lsb

some 1's, some 0's

$rax_0 = 0$

```
bsf     rcx, rax
mov     rdx, rax
dec     rdx
or      rdx, rax
mov     rax, rdx
inc     rax
mov     rbx, rdx
not     rbx
inc     rdx
and     rdx, rbx
dec     rdx
shr     rdx, cl
shr     rdx, 1
or      rax, rdx
```

$rax = (rax_0 - 1)$

$rdx = (rax_0 - 1) \vee rax_0$

? identify the zero ?

$rax = ((rax_0 - 1) \vee rax_0) + 1$

$rbx = (rax_0 - 1) \vee rax_0$

$rbx = \neg((rax_0 - 1) \vee rax_0)$

$rdx = ((rax_0 - 1) \vee rax_0) + 1$

$rdx = (((rax_0 - 1) \vee rax_0) + 1) \wedge \neg((rax_0 - 1) \vee rax_0)) - 1$

cut to first 0-byte value location ??

superficially

(3) 0011 → 0101 (5)

0011 0101

00110110

0011 1001

0011 0101

~~if finds lsb w/ an adjacent 0-bit & swaps it w/ that 0-bit~~

$rax = ((rax_0 - 1) \vee rax_0) + 1) \mid (((((rax_0 - 1) \vee rax_0) + 1) \wedge \neg((rax_0 - 1) \vee rax_0)) - 1)$

NO

if: 0110 → 1001

minimal change but related to the lsb w/ so many 1's

$[(x-1) \vee x + 1]$ "or" $[((x-1) \vee x) \wedge \neg((x-1) \vee x)] >> 1 + 1 >> *$

x OR $x-1$ & $\neg(x \text{ OR } x-1)$

first set bit of rax_0 & x

$x = 0$ ... 40

$\neg x$

$>> 1$

$\neg x$

x ... 02

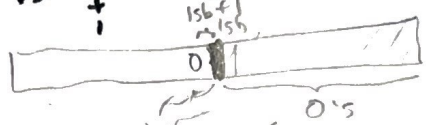
bsf rcx, rax } $rcx := $ [redacted] lsb 0-index (rax_0)

mov rdx, rax
dec rdx
or rdx, rax } $rdx = (rax_0 - 1) \vee rax_0$ } → fills every bit up to the lsb

(ex) 0010010000
0010011111

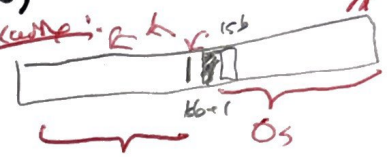
mov rax, rdx
inc rax } $rax = rdx + 1 = \overbrace{((rax_0 - 1) \vee rax_0)}^{rdx} + 1$

{NO carry =

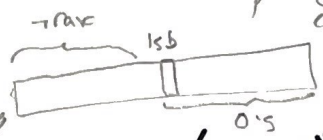


promulgates carry up if needed

{carry:

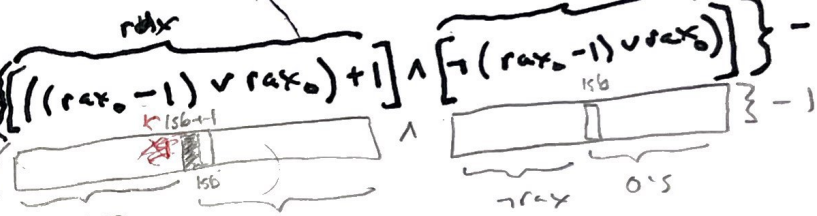


maybe propag-s ⟺ lsb+2 = 1 (cf)



mov rbx, rdx
not rbx } → $rbx = \neg rdx = \neg((rax_0 - 1) \vee rax_0)$

inc rdx } $rdx = rax_0$
and rdx, rbx
dec rdx } $rdx = \{[((rax_0 - 1) \vee rax_0) + 1] \wedge [\overbrace{\neg((rax_0 - 1) \vee rax_0)}^{rbx}]\} - 1$



$= \{$; $- 1$

$= \; - 1$

what's the diffency here?

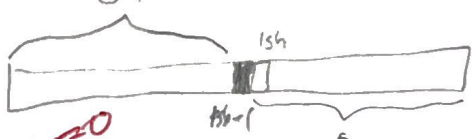
rcx
↓
shr rdx, cl
shr rdx, 1

→ $cl = bsf(rax_0)$

$rdx =$

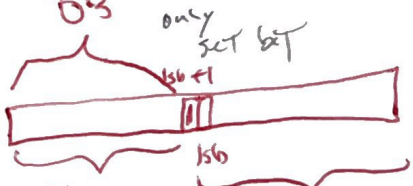
$>> 1$ → $rdx = 0 \cdots 0$

or rax, rdx



lsb+2 ≠ 0 ⇒ 0's

∴ $f(rax_0) =$



lsb+2 = 1 ⟺ potentially ≠ 0 — all 0s